Machine Learning - Undergraduate Level Assessment 20 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. What is the dimensionality of the null space of the following matrix? A = [[1, 1, 1], [1, 1, 1], [1, 1, 1]]
   (a) 0
   (b) 1
   (c) 2
   (d) 3
2. As the number of training examples goes to infinity, your model trained on that data will have:
   (a) Lower variance
   (b) Higher variance
   (c) Same variance
   (d) None of the above
3. The disadvantage of Grid search is
   (a) It can not be applied to non-differentiable functions.
   (b) It can not be applied to non-continuous functions.
   (c) It is hard to implement.
   (d) It runs reasonably slow for multiple linear regression.
4. The number of test examples needed to get statistically significant results should be _
   (a) Larger if the error rate is larger.
   (b) Larger if the error rate is smaller.
   (c) Smaller if the error rate is smaller.
   (d) It does not matter.
5. Statement 1| ImageNet has images of various resolutions. Statement 2| Caltech-101 has more images than ImageNet.
   (a) True, True
   (b) False, False
   (c) True, False
   (d) False, True

True / False (10 marks)

Answer True or False

6. True or False: Underfitting occurs when a model fails to capture the underlying patterns in the training data and cannot generalize to new data.
7. True or False: Linear hard-margin SVM can only be used when the training data are linearly separable.
8. True or False: DCGANs utilize self-attention mechanisms to enhance the stability of their training process.
9. True or False: Sampling with replacement in bagging eliminates the risk of overfitting completely for all classifiers used.
10. True or False: Calculating P(H|E, F) requires knowledge of P(E|H) and P(F|H) alone.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to count the number of inversions in the given array.
12. Write a python function to check whether the sum of divisors are same or not.

End of Paper